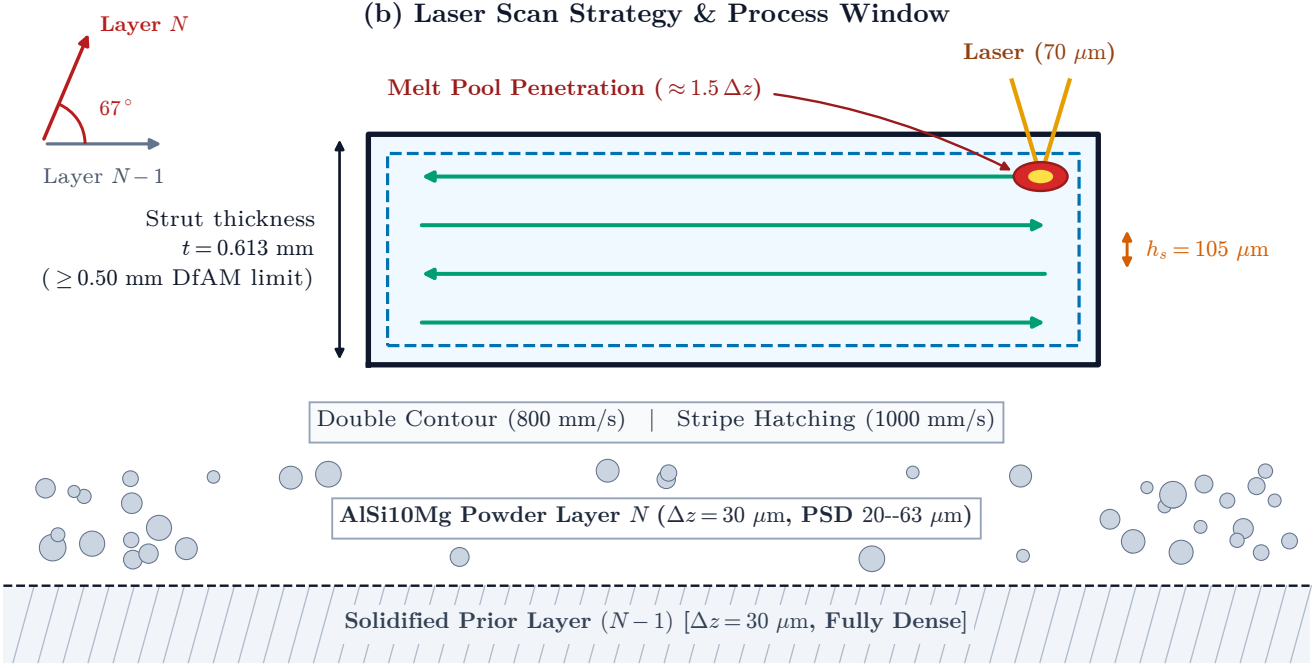

L-PBF CERTIFIED PROCESS PARAMETERS (EOS M 290 / AlSi10Mg)

Laser Exposure & Energy

- Laser Power: $P = 200$ W (Yb-fiber)
- Hatch Velocity: $v = 1000$ mm/s
- Contour Velocity: $v_c = 800$ mm/s
- Spot Diameter: $d_{\text{spot}} = 70$ μm
- Layer Height: $\Delta z = 30$ μm

Hatching & Densification

- Hatch Spacing: $h_s = 105$ μm
- Interlayer Rotation: $\Delta\alpha = 67.0^\circ$
- Volumetric Energy: $\mathbf{E_v = 63.49\ J/mm^3}$
- Relative Density: $>$ **99.8**% (Dense)
- Platform Preheat: $T_p = 150\,^\circ$C